BTS–VI–04.18–0194 Reg. No.

B

B.Tech. Degree VI Semester Examination April 2018

ME 604 HEAT AND MASS TRANSFER

(2006 Scheme)

(*Use of approved heat and mass transfer data book is permitted*)

Time: 3 Hours Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) State Fourier's law of conduction. Also give the assumptions of this law.

(b) Define efficiency and effectiveness related to fins.

(c) Briefly explain thermal boundary layer.

(d) Differentiate between pool boiling and forced convection boiling.

(e) Define intensity of radiation and state Lambert's cosine law.

(f) What is radiation shield?

(g) What are condensers and evaporators?

(h) Define fouling factor. Why is it considered for analysing heat transfer through heat exchangers?

PART B

(4 × 15 = 60)

II. (a) Derive general heat conduction equation in the Cartesian coordinates. Deduce the equation to steady state, one dimensional without heat generation. (8)

(b) A steel pipe with 50 mm OD is covered with a 6.4 mm asbestos insulation (k = 0.166 W/mK) followed by a 25 mm layer of fiber-glass insulation (k = 0.0485 W/mK). The pipe wall temperature is 393 K and the outside insulation temperature is 311 K. Calculate the interface temperature between the asbestos and fibre-glass. (7)

OR

III. (a) Derive an expression for temperature distribution for heat conduction in a plane wall with uniform heat generation when both the surfaces are at same temperature. (10)

(b) A long pipe of 0.6 m outside diameter is buried in earth with axis at a depth of 1.8 m. The surface temperatures of pipe and earth are 95°C and 25°C respectively. Calculate the heat loss from the pipe per metre length. The conductivity of earth is 0.51 W/m°C. (5)

IV. An air stream at 0°C is flowing along a heated plate at 90°C at a speed of 75 m/s. The plate is 45 cm long and 60 cm wide. Assuming the transition of boundary layer to take place at $Re_{x,c} = 5 \times 10^5$, calculate the average values of friction coefficient and heat transfer coefficient for full length of the plate. Hence calculate the rate of energy dissipation from the plate. (15)

OR

V. Calculate the convective heat loss from a radiator 0.5 m wide and 1 m high maintained at a temperature of 84°C in a room at 20°C. Treat the radiator as a vertical plate. (15)

(P.T.O.)

VI. (a) State and explain Planck's law of radiation. (5)

(b) The effective temperature of a body having an area of 0.12 m^2 is 527°C. Calculate the following. (10)

(i) The total rate of energy emission.

(ii) The intensity of normal radiation.

(iii) The wavelength of maximum monochromatic emissive power.

OR

VII. (a) Emissivity of two large parallel plates maintained at 800°C and 300°C are 0.3 and 0.5 respectively. Find the net radiant heat exchange per square metre for these plates. (7)

(b) A boiler furnace lagged with plate steel is lined with fireclay bricks on the inside. The temperature of the outside of the brick setting is 127°C and the temperature of the inside of the steel plate is 50°C. Assuming the gap between the plate steel and fireclay bricks to be small compared with the size of the furnace, calculate the loss of heat per unit area by radiation between the lagging and setting (Emissivity for steel = 0.6, emissivity for fireclay = 0.8). (8)

VIII. (a) How are heat exchangers classified? (5)

(b) Derive an expression for effectiveness of a parallel flow heat exchanger. (10)

OR

IX. A 1 shell-2 pass steam condenser consists of 3000 brass tubes of 20 mm diameter. Cooling water enters the tubes at 20°C with a mean flow rate of 3000 kg/s. The heat transfer coefficient for condensation on the outer surfaces of the tubes is 15500 W/m^2K. If the heat load of the condenser is 2.3×10^8 W, when the steam condenses at 50°C, determine: (15)

(i) The outlet temperature of the cooling water.

(ii) The overall heat transfer coefficient.

(iii) The tube length per pass.

(iv) The rate of condensation of steam if h_{fg} = 2380 kJ/kg.

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